

PRACTICAL WEEK 2

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MySQL 8.0 Command Line Cli × + v

Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE prac2;
Query OK, 1 row affected (0.01 sec)

mysql> USE prac2;
Database changed
mysql> CREATE TABLE Books (
  -> book_id INT PRIMARY KEY,
  -> title VARCHAR(100) NOT NULL,
  -> isbn VARCHAR(20) UNIQUE,
  -> published_year INT CHECK (published_year < 2027)
  -> );
Query OK, 0 rows affected (0.08 sec)

mysql>
mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES
  -> (1, 'The Great Gatsby', '9780743273565', 1925),
  -> (2, 'To Kill a Mockingbird', '9780061120084', 1960),
  -> (3, '1984', '9780451524935', 1949);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title                | isbn          | published_year |
+-----+-----+-----+-----+
| 1       | The Great Gatsby     | 9780743273565 | 1925           |
| 2       | To Kill a Mockingbird | 9780061120084 | 1960           |
| 3       | 1984                 | 9780451524935 | 1949           |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
mysql> CREATE TABLE Members (
  -> member_id INT PRIMARY KEY,
  -> full_name VARCHAR(100),
  -> email VARCHAR(100) UNIQUE
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> INSERT INTO Members (member_id, full_name, email) VALUES
  -> (101, 'John Smith', 'john.smith@email.com'),
  -> (102, 'Emma Wilson', 'emma.wilson@email.com'),
  -> (103, 'Michael Brown', 'michael.brown@email.com');
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0
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mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name      | email                      |
+-----+-----+-----+
| 101       | John Smith     | john.smith@email.com      |
| 102       | Emma Wilson    | emma.wilson@email.com     |
| 103       | Michael Brown  | michael.brown@email.com   |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
mysql> CREATE TABLE Loans (
  -> loan_id INT PRIMARY KEY,
  -> member_id INT,
  -> book_id INT,
  -> loan_date DATE,
  -> FOREIGN KEY (member_id) REFERENCES Members(member_id),
  -> FOREIGN KEY (book_id) REFERENCES Books(book_id)
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
  -> (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
  -> (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
  -> (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
  -> (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
  -> (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql>
mysql> SELECT m.full_name AS Member_Name, b.title AS Book_Title
  -> FROM Loans l
  -> INNER JOIN Members m ON l.member_id = m.member_id
  -> INNER JOIN Books b ON l.book_id = b.book_id;
+-----+-----+
| Member_Name | Book_Title |
+-----+-----+
| John Smith  | The Great Gatsby |
| John Smith  | To Kill a Mockingbird |
| John Smith  | 1984 |
| John Smith  | The Great Gatsby |
| Emma Wilson | To Kill a Mockingbird |
| Emma Wilson | The Great Gatsby |
| Emma Wilson | 1984 |
| Michael Brown | 1984 |
| Michael Brown | To Kill a Mockingbird |
| Michael Brown | The Great Gatsby |
+-----+-----+
10 rows in set (0.00 sec)

mysql>
mysql> SELECT published_year, COUNT(book_id) AS Total_Books
  -> FROM Books
  -> GROUP BY published_year
  -> ORDER BY published_year;
+-----+-----+
```

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+-----+-----+
10 rows in set (0.00 sec)

mysql>
mysql> SELECT published_year, COUNT(book_id) AS Total_Books
    -> FROM Books
    -> GROUP BY published_year
    -> ORDER BY published_year;
+-----+-----+
| published_year | Total_Books |
+-----+-----+
|          1925 |          1 |
|          1949 |          1 |
|          1960 |          1 |
+-----+-----+
3 rows in set (0.00 sec)

mysql>
mysql> CREATE TABLE Donation_History (
    -> donation_id INT PRIMARY KEY,
    -> book_id INT,
    -> donor_name VARCHAR(100),
    -> donation_date DATE,
    -> FOREIGN KEY (book_id) REFERENCES Books(book_id)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO Books (book_id, title, isbn, published_year)
    -> VALUES (4, 'Animal Farm', '9780451526342', 1945);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
    -> VALUES (1, 4, 'Raj Kumar', CURDATE());
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> CREATE INDEX idx_books_isbn ON Books(isbn);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Books WHERE isbn = '9780451524935';
+-----+-----+-----+-----+
| book_id | title | isbn          | published_year |
+-----+-----+-----+-----+
|        3 | 1984  | 9780451524935 |          1949 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql>
```

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CODE:

```
CREATE DATABASE prac2;
USE prac2;
CREATE TABLE Books (
    book_id INT PRIMARY KEY,
    title VARCHAR(100) NOT NULL,
    isbn VARCHAR(20) UNIQUE,
    published_year INT CHECK (published_year < 2027)
);

INSERT INTO Books (book_id, title, isbn, published_year) VALUES
(1, 'The Great Gatsby', '9780743273565', 1925),
(2, 'To Kill a Mockingbird', '9780061120084', 1960),
(3, '1984', '9780451524935', 1949);
SELECT * FROM Books;

CREATE TABLE Members (
    member_id INT PRIMARY KEY,
    full_name VARCHAR(100),
    email VARCHAR(100) UNIQUE
);

INSERT INTO Members (member_id, full_name, email) VALUES
(101, 'John Smith', 'john.smith@email.com'),
(102, 'Emma Wilson', 'emma.wilson@email.com'),
(103, 'Michael Brown', 'michael.brown@email.com');
SELECT * FROM Members;

CREATE TABLE Loans (
    loan_id INT PRIMARY KEY,
    member_id INT,
    book_id INT,
    loan_date DATE,
    FOREIGN KEY (member_id) REFERENCES Members(member_id),
    FOREIGN KEY (book_id) REFERENCES Books(book_id)
);

INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
(1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
(3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
(5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
(7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
(9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');

SELECT m.full_name AS Member_Name, b.title AS Book_Title
FROM Loans l
INNER JOIN Members m ON l.member_id = m.member_id
INNER JOIN Books b ON l.book_id = b.book_id;

SELECT published_year, COUNT(book_id) AS Total_Books
FROM Books
GROUP BY published_year
ORDER BY published_year;

CREATE TABLE Donation_History (
    donation_id INT PRIMARY KEY,
    book_id INT,
    donor_name VARCHAR(100),
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donation_date DATE,  
FOREIGN KEY (book_id) REFERENCES Books(book_id)  
);
```

```
START TRANSACTION;
```

```
INSERT INTO Books (book_id, title, isbn, published_year)  
VALUES (4, 'Animal Farm', '9780451526342', 1945);
```

```
INSERT INTO Donation_History (donation_id, book_id, donor_name,  
donation_date)  
VALUES (1, 4, 'Raj Kumar', CURDATE());
```

```
COMMIT;
```

```
CREATE INDEX idx_books_isbn ON Books(isbn);  
SELECT * FROM Books WHERE isbn = '9780451524935';
```